

Ambient Gantt

Leveraging Ambient Biometrics and Morale Insights for Optimal Front-Line Deployment

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Inspired by: Philips RATE Program & Current VA Project

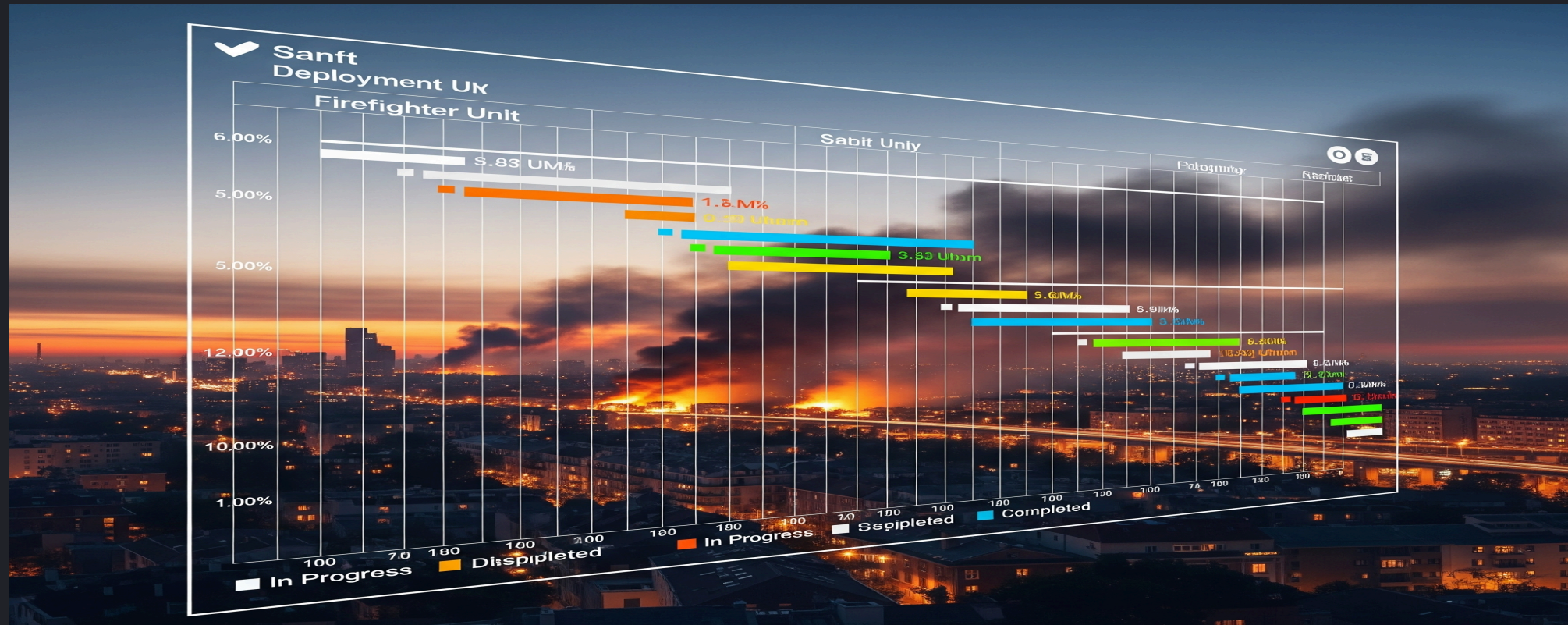
1. The Human Factor: Why We're Here

- **The Compelling Event:** There's a silver-lining from COVID: focused data pipeline dev towards predicting ONE health condition...
- **Today's Paradigm:** Manual schedules, self-reported fitness, subjective assessments.
- **The Hidden Threats:** Stress, fatigue, and nascent illness are invisible.
- **The Vision:** Proactively support our personnel with a real-time understanding of their well-being.

2. The Technology: Seeing the Invisible

- **Patient-Generated Health Data (PGHD):** Data from wearables (watches, rings) and smart devices.
- **Ambient Tracking:** Passive, continuous monitoring of key biometrics.
 - **Sleep:** Quality, duration, disturbances.
 - **Respiratory Health:** Rate, heart rate variability (HRV), even airborne compounds to detect early signs of illness.
 - **Stress & Fatigue**

3. The Application: A "Readiness" Gantt Chart



4. How an "Interchange" Works

- **Alert:** Firefighter John Doe's data flags a high probability of a developing respiratory illness. His status turns **Red**.
 - **Action:** The resource manager is alerted and seamlessly exchanges him with a **Green** status firefighter on standby.
 - **Support:** John Doe is directed to health services for early intervention.
- The Result:** The crew is protected, the mission is fully staffed, and John receives care before his condition

5. The Fuzzy Area: Quantifying Morale

- **Beyond Physical Health:** Low morale can precede burnout, errors, and risk.
- **Measuring the "Unmeasurable":**
 - Anonymous, brief pulse surveys (e.g., "Rate your morale today: 1-5").
 - Analyzing anonymized team-level sentiment.
 - Tracking engagement in peer support and training.
- **The Goal:** Not to punish, but to provide a "psychological pit stop."

6.1 The Ethical Minefield & The Path Forward

- **Foundation of Trust is NON-NEGOTIABLE.**
 - **Data Privacy:** Anonymized, secure, and compliant with HIPAA, ADA, etc.
 - **Transparency & Consent:** Must be opt-in, with clear understanding of use.
 - **Governance:** A multi-stakeholder committee with union and ethics oversight.
- **This is a Support Tool, NOT a Surveillance Tool.**

6.2 The Ethical Minefield & The Path Forward

- **Socioeconomic REALITY**
 - **Work Ethic:** Diverse work ethics should be considered.
 - **Sacrifice:** A single-mother nurse who is told she can't work?!
 - **Agency:** Free-will for the individual in civilian world.

7. Conclusion & Call to Action

- **The Vision:** Empower our heroes with data-driven, proactive care.
- **The Choice:** Remain with the limitations of today or build the supportive, resilient future of tomorrow.
- **The Ask:** Brainstorm resource-allocation informed by biometrics for a **commercial** application.

Q&A

Thank You